

TAMIL NADU STATE BOARD - STANDARD 12 CHEMISTRY VOLUME 2

CHAPTER 14: BIOMOLECULES

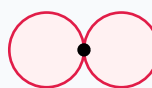
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Biomolecules

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)



p orbital (px)

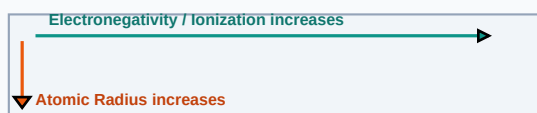
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Zwitterion	An amino acid dipolar ion containing equal numbers of positively and negatively charged functional groups, showing net neutrality.
Peptide Bond	A covalent amide linkage (-CO-NH-) formed by condensation of carboxyl and amino groups of two amino acids.
Denaturation of Proteins	The disruption of secondary, tertiary, and quaternary protein structures by heat or pH shifts, losing biological activity.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

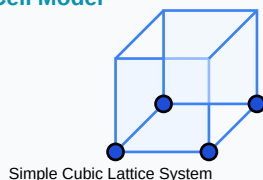
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Mutarotation: Alpha-D-glucose (112 deg) and Beta-D-glucose (19 deg) equilibrate to 52.7 deg
- Nucleotide: Pentose sugar + Nitrogenous base + Phosphate

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

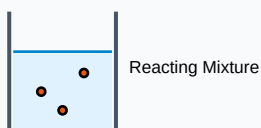
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Differentiate between DNA and RNA regarding sugar and bases.

Step-by-step Solution:

DNA contains deoxyribose sugar and thymine base (A, T, G, C). RNA contains ribose sugar and uracil base (A, U, G, C).

Titration Beaker & Solute Precipitation



HOTS (Higher Order Thinking Skills) Challenge

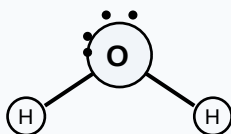
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Describe the structural features of DNA double helix as proposed by Watson and Crick.

Analytical Solution Strategy:

Consists of two polynucleotide chains coiled around a common axis. The chains run anti-parallel (5'→3' and 3'→5'). Bases are inside, sugars outside. Adenine pairs with Thymine (2 H-bonds); Guanine pairs with Cytosine (3 H-bonds).

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

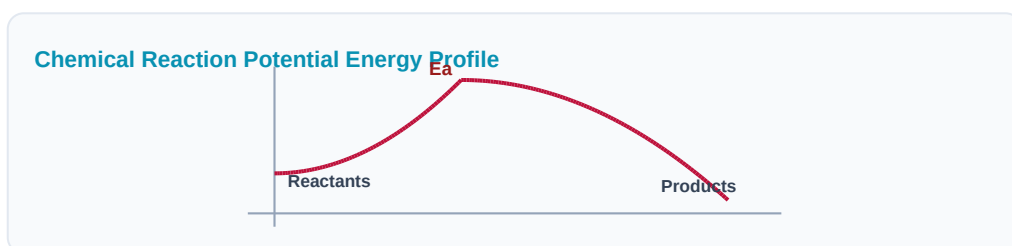
Q1 (1-Mark): Explain mutarotation of glucose.

Q2 (2-Mark): Describe the structure of amino acids. What is isoelectric point?

Q3 (2-Mark): Compare fibrous proteins and globular proteins.

Q4 (5-Mark): Explain the primary, secondary, and tertiary structures of proteins.

Q5 (5-Mark): What are vitamins? Classify them based on solubility with examples.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Food Nutrition

Starch carbohydrate polymers are metabolized into blood glucose to fuel physical activity.

Application Case: Forensic Science

DNA double helix stability allows matching genetic signatures in crime assays.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

